

EEG Signal Classification in Neurological Disorders

Abstract

Electroencephalogram (EEG) signals provide valuable insights into brain activity and are widely used for diagnosing neurological and mental health disorders. However, manual interpretation of EEG data is complex, time-consuming, and prone to human error. This project proposes an intelligent and secure EEG signal classification system using a deep learning-based Long Short-Term Memory (LSTM) network to automatically identify multiple neurological disorders, including Anxiety Disorder, Mood Disorder, Obsessive-Compulsive Disorder (OCD), Schizophrenia, Trauma and Stress-related Disorder, Addictive Disorder, and healthy conditions. The system preprocesses raw EEG signals through filtering, normalization, and segmentation to enhance data quality before feeding them into the LSTM model for time-series learning and multi-class classification. To address the lack of transparency in deep learning models, an LLM-based explanation module is integrated to generate humanreadable interpretations of predictions, improving trust and clinical usability. Furthermore, a private blockchain implemented using Ganache ensures secure, tamper-proof storage of classification results, preserving data integrity and transparency. A web-based client-server architecture developed using Flask, HTML, and CSS enables real-time EEG data upload, prediction visualization, explanation generation, and blockchain transaction tracking. Experimental results demonstrate that the proposed LSTM-based approach achieves high accuracy and outperforms traditional machine learning models in EEG classification tasks. Overall, this system combines deep learning, explainable AI, and blockchain technology to deliver a reliable, transparent, and secure framework for automated neurological disorder detection, offering strong potential for clinical decision support and academic research.

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